

WARESAFE

A SECURE IOT-BASED FACTORY AND WAREHOUSE SECURITY SYSTEM WITH CYBERSECURITY SIMULATION

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CLIENT: Farah Consulting Tech – Lmesh BV

01

EXISTING CONDITION

- **IoT** adoption in factories and warehouses is **increasing**.
- Industrial IoT systems are **vulnerable** to **unauthorized access, data manipulation**, and **DDoS attacks**.
- Manual monitoring: patrols, recording, and detection can cause **delays for hours**.

02

CORE PROBLEMS

- **Testing** cyber attacks on **live production** systems is **risky**.
- The **absence** of an **isolated testing environment**.
- Physical monitoring and cybersecurity testing are **not separated**.
- Conducting **realistic security evaluations** remains challenging

03

SUGGESTED IMPROVEMENT

- IoT sensors are **interconnected**.
- Data recording is **fully digital** and **automated**.
- **Sensors, communication, dashboards**, and **real-time alerts** are **integrated**.
- A dedicated **cybersecurity simulation layer**.
- **No duplicate hardware** is required.

WARESAFE ARCHITECTURE

01

Physical Layer

Sensors: IR Motion Sensor, RFID Card, Door Reed Switch

02

Edge Processing Layer

ESP32 FT232

03

Communication Layer

MQTT (Eclipse Mosquitto)

04

Application Layer

Laravel, MySQL, Unity

05

Presentation Layer

Dashboard and 2D Game Simulation

DEVICES AND SENSORS PLACEMENT

Office — 1st Floor

IR Motion Sensor
Door Reed Switch
Vibration
Buzzer as Alert

Warehouse — 2nd Floor

RFID Card
Door Reed Switch
LCD Display as Alert
LED Light as Alert
Buzzer as Alert

CYBERSECURITY SCENARIO

Office — 1st Floor

Replay Attack

Vibration
Spoofing

Warehouse — 2nd Floor

False Alarm
Flooding

Security
Supression

+ Defense Mechanism

2D GAME SIMULATION

Key Features

- Hands-on cybersecurity learning experience
- Explore 2 zones, 4 cybersecurity scenarios
- Simple game controls
- Real-time connect to MQTT
- Choices-based quiz

DASHBOARD MONITORING

Key Features

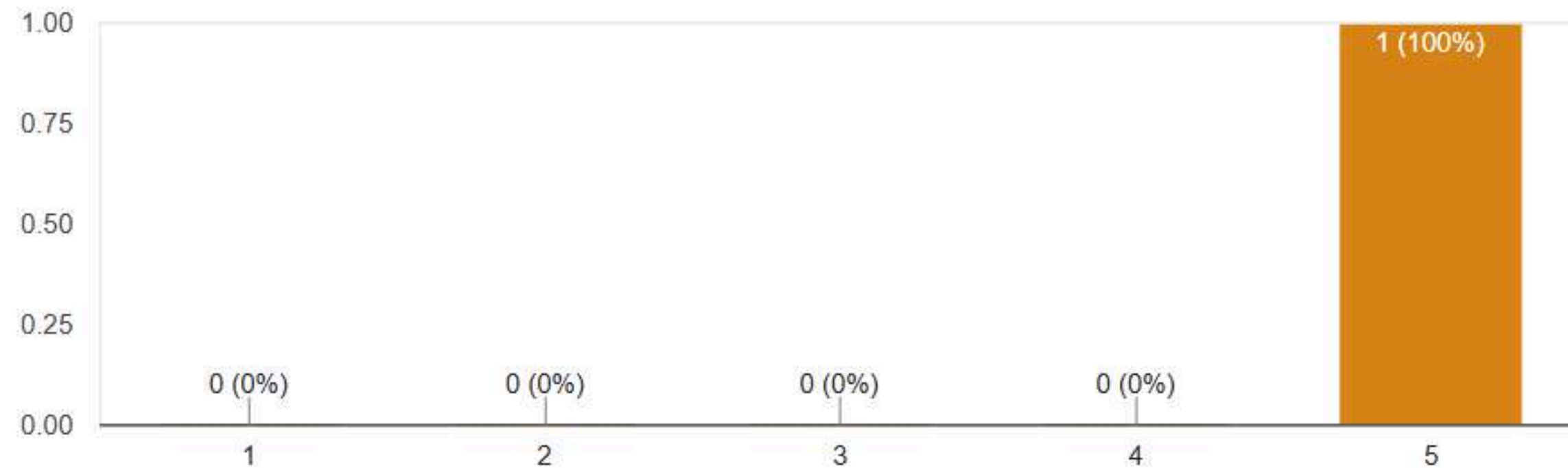
- Overview Page
- Zone, Device, Sensor, and MQTT topics Management
- Sensor Monitoring
- Activity Records
- Mitigation Education Module
- Simple interface

USER FEEDBACK

Does WareSafe meet the Client's expectations as a smart warehouse cybersecurity training and monitoring solution?

 Copy chart

1 response





LIVE DEMO



THANK
YOU

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